An Adaptive Multiview Active Learning Approach for Spectral–Spatial Classification of Hyperspectral Images

Zhou Zhang, *Member, IEEE*, Edoardo Pasolli, *Senior Member, IEEE*, and Melba M. Crawford, *Fellow, IEEE*

***Abstract*— Combining spectral and spatial features in hyperspectral image classification is a common practice due to the improvements in classification accuracy that can be obtained by extracting information from neighboring pixels. However, the resulting high dimensionality of the input data and the typically limited number of labeled samples are two key challenges that affect the overall performance of supervised classification methods. To alleviate these two issues, we propose an adaptive multiview (MV)-based active learning (AL) approach that is different from the existing MV AL methods in two main ways: 1) to improve the view sufficiency, a spectral–spatial view generation approach is proposed by incorporating spatial features derived from the segmentation maps into each view and 2) to increase the diversity across views, a dynamic view is generated at each AL iteration by selecting important features from the predefined views. The performance of each view is further improved by applying the proposed AL algorithm in conjunction with an ensemble approach as back-end classifier, a scenario less explored in the remote sensing community than single classifier-based AL methodologies. The proposed approach is applied to three widely analyzed hyperspectral data sets [i.e., Kennedy Space Center (KSC), Indian Pine, and University of Houston (UH)], and the results demonstrate the efficacy of the proposed method compared with other state-of-the-art AL classification methods.**

***Index Terms*— Active learning (AL), classification, dynamic view, hyperspectral data, multiview (MV), spatial features.**

I. Introduction

HYPERSPECTRAL sensors can collect data in hundreds of narrow, contiguous bands, thus providing comprehensive spectral information over a range of values. Land cover mapping is the primary application of hyperspectral imagery since the detailed spectral information is quite useful for identifying and discriminating different land cover types. Besides the spectral information, spatial information derived from the neighboring pixels is frequently incorporated into classification models, since integration of disparate features often provides valuable complementary information that can potentially increase the classification accuracy [1]–[5]. Supervised classification approaches are often used to generate an accurate land cover map. The performance of a supervised algorithm strongly depends on the "quality" and "quantity" of training samples. However, collecting training data is expensive and time-consuming, and these data are typically collected on a limited portion of the image. In such cases, the training samples may not be representative of the statistical distribution of the data, thus resulting in degraded performance of a supervised classifier. Active learning (AL), an iterative procedure of selecting the most informative samples to add into the training set, provides a promising way to address this issue [6]. More specifically, the uncertainty of a pixel's class label is typically used to quantify its informativeness. This is because using the samples with maximum uncertainty in their class labels for training can further improve the classification model.

AL approaches have been successfully applied in remote sensing image classification, and most approaches focus on the spectral domain. In general, these studies can be grouped into three main classes: 1) margin sampling (MS)-based approaches, where the samples closest to the separating hyperplane of the classifier, such as support vector machines (SVMs), are considered as the most uncertain ones [7]; 2) committee learner-based approaches, where the ones exhibiting maximal disagreement between the different classification models are selected [8]; and 3) class probability distribution-based approaches, and breaking ties (BT) where the difference between the two highest posterior probabilities is used to quantify the uncertainty of a pixel [9]. For AL algorithms developed in the spectral domain, we refer the reader to [10] and [11] for a survey for AL.

Considering potential improvements that can be obtained by incorporating spatial features, AL approaches that use both spectral and spatial information have been studied recently and can be categorized into two main approaches based on how the spatial information is introduced. In the first type of approach, spatial features are incorporated directly during the classification process. The spatial features are either combined with the spectral features to form an enriched input feature space or used to regularize the classification map [12]–[16]. In the second type of approach, the spatial information is used

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Z. Zhang is with the Department of Biological Systems Engineering, University of Wisconsin–Madison, Madison, WI 53706 USA (e-mail: zzhang347@wisc.edu).

E. Pasolli is with the Department of Agricultural Sciences, University of Naples Federico II, 80055 Naples, Italy (e-mail: edoardo.pasolli@unina.it).

M. M. Crawford is with the School of Civil Engineering, Purdue University, West Lafayette, IN 47907 USA (e-mail: mcrawford@purdue.edu).

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for designing the AL query strategy, but the classification is still performed in the spectral domain [17]–[20]. Different AL query criteria have been proposed by incorporating the spatial information. Some strategies focus on selecting the samples that are informative in both spectral and spatial domains [17], [18] and others use the spatial information for reducing the labeling cost by considering the traveling distance during the field survey [19], [20]. There are also some approaches that combine these two strategies into one framework [21], [22].

In the remote sensing community, multiview (MV)-based AL methods, which belong to the committee-based AL category, have received significant attention, especially when the disparate features (e.g., spectral, spatial, and vegetation structural information) are combined for classification. In contrast to the single-view (SV) scenario in which a learner has access to all the features, an MV setting partitions the entire feature space into different views (subspaces) that are sufficient for learning the target function and diverse for providing complementary information. Each view trains a classifier independently, and the final classification map is obtained by fusing the results from the multiple learners. Compared with SV AL, an MV AL framework typically has three main advantages [21], [23]. First, it requires fewer training samples to get the same accuracy due to the lower dimensionality of each SV. Second, it can take advantage of the complementary information provided by the different views. Finally, it has a simple, flexible formulation since no assumptions are needed about the properties of the base learners.

Motivated by the success of spectral–spatial classification and the effectiveness of MV AL, we propose, in this article, an adaptive MV AL approach for hyperspectral image classification with two main contributions: 1) both spectral and spatial features are incorporated for view generation with spatial features derived from segmentation maps and 2) a dynamic view is generated at each AL iteration by selecting the important features from the predefined fixed views. We further adopted an ensemble approach as back-end classifier to further improve the performance of each view. This represents a scenario less investigated in the remote sensing community than AL methodologies based on single classifiers. The rest of this article is arranged as follows. The spatial feature extraction approach and the MV AL framework are introduced in Section II. In Section III, the proposed adaptive MV AL method for hyperspectral image classification is described. Data sets used in the experimental analysis and the obtained experimental results are presented in Section IV. Conclusions are summarized in Section V.

II. Related Work

A. Hierarchical Segmentation for Spatial Feature Extraction

In general, inclusion of spatial information in classification of image data is directly related to the definition of a pixel neighborhood. A fixed-size window is usually adopted to define the neighborhood, and spatial features can be easily extracted from it. However, this method suffers from the so-called "border effect" problem where the neighborhood includes pixels from multiple objects. To address this problem, the neighborhood can be defined adaptively via segmentation. Segmentation methods partition an image into nonoverlapping homogeneous regions using a defined homogeneity criterion; each resulting region in a segmentation map defines a spatial neighborhood for all the pixels within this region. A wide range of methods for segmentation of multispectral and hyperspectral images have been introduced in the literature. Some exploit mathematical morphology-based segmentation approaches, which mostly use watershed transformations [24], [25], while others exploit clustering-based segmentation approaches [26] and multiscale techniques [27]. Hierarchical methods have also been studied in order to consider the spatial constraint during the clustering procedure [28], [29]. In this context, the hierarchical segmentation (HSeg) algorithm, which naturally integrates spatial and spectral information, has recently received much attention [30], [31].

HSeg [31] combines region growing, which produces spatially connected regions, with clustering, which groups spatially disjoint but spectrally similar regions. The segmentation is initialized by assigning a region label to each image pixel. Then, the HSeg partitions the image iteratively by merging the regions with minimum dissimilarity until the entire image is assigned to a single region. The merging process depends on two factors: the dissimilarity criterion (DC) and the parameter *S_wght*, which sets the importance of spatially adjacent regions with respect to nonadjacent ones. Since the output of the HSeg is a segmentation hierarchy, which contains different levels of detail of the image, it is necessary to define an optimal segmentation from the hierarchy. Several approaches have been proposed to find the best segmentation. In [30] and [31], the best segmentation is selected from one level of the hierarchy, while in [22] and [32], the best segmentation is constructed by selecting regions at different levels. To generate the spectral–spatial views, in this article, we apply the supervised pruning strategy proposed in [22] to obtain the best segmentation map from the outputs of HSeg, as it has been demonstrated to perform better than the other approaches to determine the hierarchy [22].

B. MV AL for Hyperspectral Image Classification

In the machine learning community, the co-training [33] algorithm, which works in a semisupervised manner, provides the basis for MV learning. Following co-training, the first MV AL algorithm was co-testing [23], a two-step iterative scheme that: 1) trains a learner in each view using the available training samples and 2) queries samples on which the views have different label predictions. When the views disagree, at least one of them is wrong. Therefore, such queries are highly informative as they correct mistakes made by at least one of the views. These approaches and their extensions, including co-expectation maximization (Co-EM) [34], Co-EM+Co-Testing (Co-EMT) [35], and Co-EM SVM [36], have been successfully applied in many fields, such as web page classification and text categorization. To ensure their success, views that are both sufficient and diverse are required so that each view is capable

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of learning the target function, and the views are unlikely to make the same mistakes [23].

In this article, we focus on MV AL for hyperspectral image classification. In a hyperspectral image, let $\mathbf{x}_i \in \mathbf{R}^b$ denote a pixel in the image with b spectral bands, and $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{n}$ denote the whole image with n pixels. For supervised classification, let $L = \{\mathbf{x}_i, y_i\}_{i=1}^{l}$ be a set of training samples, which is constructed by selecting l samples from the image $\mathbf{X}$, with corresponding class labels $y_i \in \{1, 2, \ldots, W\}$, where W is the number of classes, and $U = \{\mathbf{x}_i\}_{i=l+1}^{l+u}$ be the set of the remaining unlabeled samples, where u denotes the number of unlabeled pixels, and we have $U = X\text{-}L$ and $n = l+u$. The classification map is generated by assigning one label from the W classes to each pixel in the image. A general MV AL framework typically requires four components that are given as follows.

1) *View Generation:* Identification of J different views from the original data to provide the input to the multiple classifiers, and representation of a pixel $\mathbf{x}_i$ is $(\mathbf{x}_i^{V_1}, \mathbf{x}_i^{V_2}, \ldots, \mathbf{x}_i^{V_J})$.
2) *Back-End Classifier:* Train a classifier for each view with the labeled data separately. Then, apply the trained classifier $\{\mathbf{C}_1, \mathbf{C}_2, \ldots, \mathbf{C}_J\}$ to the unlabeled sample set U, and obtain J predictions for each pixel.
3) *AL Query Criterion:* At each AL iteration, select the ba most informative samples from set U with a query strategy. Add the ba samples into L and remove them from U.
4) *Decision Fusion Strategy*: The final classification map is obtained by combining all the J predictions from $\{\mathbf{C}_1, \mathbf{C}_2, \ldots, \mathbf{C}_J\}$ via a fusion strategy.

In the current literature, several MV AL approaches have been developed for hyperspectral data classification, and they differ in one or more of the abovementioned four components. In [37], several view generation methods are introduced for MV AL for hyperspectral image classification, including clustering, uniform slicing, and random generation, and a maximum disagreement-based AL query strategy is used for selecting the most uncertain samples. Based on [37], a coregularization AL query approach is proposed in [38]. Manifold learning in both the spectral and spatial domains is incorporated into the AL process to further select the most informative samples into the training set. To effectively combine multiple types of features, an ensemble multiple kernel learning AL approach is explored in [39], in which each view represents one feature type (source sensor, spatial, and spectral), and thus satisfies the necessary diversity condition for view construction. In [21], a wavelet-domain MV AL framework is developed for hyperspectral image classification. A 3-D redundant wavelet transform (3D-RDWT) is used to generate MVs that are less sensitive to additive noise. Recently, an intensity-based MV AL approach is proposed in which MVs are generated by utilizing multiple morphological component analysis (MMCA) to provide good diversity and complementarity, and the independent prediction ability of each view is investigated for the AL query [40].

Although encouraging results were obtained in the existing MV AL frameworks, the following drawbacks still remain.

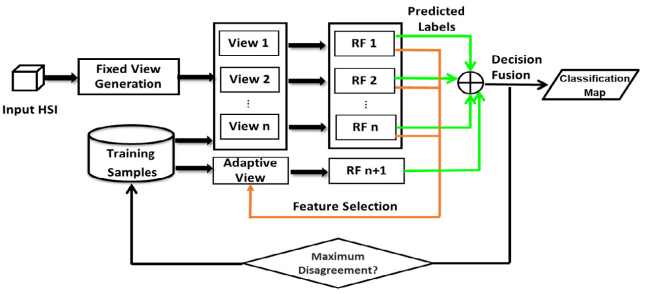


Fig. 1. Flowchart of the proposed MV AL framework. RF: random forest classifier.

First, only one type of feature is considered in each view, which limits the view sufficiency condition. Second, the views are mostly fixed during the AL process. A drawback of using fixed views in AL is that views tend to agree with each other as AL proceeds and thus degrade the performance of the AL query strategy, which typically relies on the disagreement level across views [37]. Although some view updating approaches were investigated in [37], such as increasing the number of views and switching to different view generating approaches during the AL process, all the views need to be regenerated for updating, which is quite time consuming, especially when more than one type of feature is considered in each view. Third, all the approaches proposed in the literature adopt a single classifier, e.g. SVM, as the back-end classifier, although an ensemble back-end classifier has the potential to further improve the view sufficiency. To alleviate these problems, we propose a new MV AL approach whose details are described in Section III.

III. Proposed Method

The overall flowchart of the proposed MV AL framework is summarized in Fig. 1. First, we generate the fixed views by using both spectral and spatial features derived from HSeg outside the AL loop. Then, in the online AL part of the method, a random forest (RF) classifier is trained for each fixed view and the feature importance (FI) is calculated accordingly. Given the FI, a dynamic view is constructed by selecting a certain number of important features from each fixed view. To enrich the training set by AL, at each iteration, the unlabeled samples are ranked, and the most uncertain ones are selected, labeled, and added to the training set. The classification results are obtained by combining the predictions from all the views using a linear opinion pool (LOP) strategy. The whole procedure is repeated until a predefined number of training samples is reached. The proposed MV AL framework is summarized as follows, and the details of each component are discussed in Sections III-A–III-D.

A. Fixed View Generation

It is important to generate appropriate views to conduct a successful MV AL. In view generation, two main conditions must be satisfied: sufficiency and diversity [37]. This means that each view should possess good discrimination capabilities, but at the same time, the dependence between the views

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Algorithm 1 Proposed MV AL Framework

Initialization:

- Initial labeled training set $L = \{\mathbf{x}_i, y_i\}_{i=1}^{l}$
- Initial unlabeled learning set $U = \{\mathbf{x}_i\}_{i=l+1}^{l+u}$
- Generate J-1 fixed spectral-spatial views $\{V_1, V_2 \ldots, V_{J-1}\}$ by extracting spatial features from HSeg
- Number of important features selected from each fixed view q
- AL batch size ba, i.e. the number of samples to select and label at each AL step
- Total number of final training samples S

Main AL Loop:

- Train J-1 RF classifiers $\{\mathbf{C}_1, \mathbf{C}_2 \ldots, \mathbf{C}_{J-1}\}$ on the fixed views $\{V_1, V_2 \ldots, V_{J-1}\}$
- Calculate the feature importance for all the fixed views
- Generate the dynamic view V_J by selecting q important features from each fixed view
- Train a RF classifier $\mathbf{C}_J$ on the dynamic view V_J
- Select ba samples from U based on the maximum disagreement query strategy
- Add the ba samples to the training set L after labeling and remove them from the learning set U
- Obtain the classification map by combing the predictions from the J views based on LOP

Stop when the training set L contains S samples

must be minimized for providing complementary information. We propose to accomplish this by adopting the correlation-partition-based clustering approach. This strategy, which also implicitly determines the number of views, was introduced in [37] to partition the original hyperspectral space (i.e., without considering any spatial information) into multiple disjoint contiguous views. Diversity can be satisfied since each view has high intraband correlation but low correlation with the other views, indicating that the complementary information can be introduced by different views. However, this method suffers from the risk of insufficiency since using spectral information alone may not provide adequate information for each view. To address this issue, we propose to incorporate spatial features into each view in order to improve the sufficiency condition of each view. Using a fixed-size window is a common strategy to extract the spatial features for each view, which, however, suffers from the "border effect" problem. In addition, using a fixed size window for all the views implies to consider the same neighborhood for all the spectral bands, which limits the advantage of the MV scenario approach. Therefore, it is preferable to define a spatial feature extraction process in which the neighborhood is view-specific in order to improve both sufficiency and diversity across different views.

To add diverse spatial features into each view, we used the HSeg algorithm combined with the supervised tree pruning strategy [22] for spatial feature extraction in which the spatial features are calculated for each region in the derived best segmentation map. Based on this, the spectral–spatial view generation procedure includes four steps: 1) generating spectral views for the original hyperspectral data based on the correlation-partition-based clustering approach [37]; 2) running HSeg for each spectral view separately; 3) applying the supervised pruning strategy with the initial training data to obtain the best segmentation map for each view [22]; and 4) extracting the spatial features for each view, which are the first two moments (i.e., mean and standard deviation), calculated for each region in the best segmentation map. Specifically, for each view, the mean and standard deviation are calculated for each band in that view based on the segmentation map. Assume that the sample $\mathbf{x}_i$ is in the region q of the segmentation, and the region has in total N_q pixels. The mean and standard deviation for the sample $\mathbf{x}_i$ at the bth band of view V_j can be calculated as

$$\mathbf{x}_i^{V_j^b}\text{_spaMean} = \frac{\sum_{t=1}^{N_q} \mathbf{x}_t^{V_j^b}}{N_q}$$

$$\mathbf{x}_i^{V_j^b}\text{_spaStd} = \sqrt{\frac{\sum_{t=1}^{N_q} \left(\mathbf{x}_t^{V_j^b} - \mathbf{x}_i^{V_j^b}\text{_spaMean}\right)^2}{N_q - 1}}.$$

The extracted spatial features will then be concatenated with the spectral features in order to form the final spectral–spatial feature set. It is important to note that for each pixel, unlike the fixed-size window, the derived neighborhoods from the segmentations are different across different views. Therefore, the extracted spatial features are view-specific and the view diversity is potentially increased by considering different neighborhoods for each pixel.

B. Ensemble Back-End Classifier

In general, AL hyperspectral image classification problems have been studied from three aspects: SV single classifier (SVSC), SV ensemble classifier (SVEnC), and MV single classifier (MVSC). Among these three approaches, SVSC is the classic method that researchers have utilized in AL problems, and some AL baseline methods, such as MS and BT, use this strategy in conjunction with the SVM classifier [12]. The SVEnC trains ensemble classifiers on the same feature set. When it is used for AL, for an unlabeled sample, committees would have different predictions, and the sample with the most disagreement is selected. A representative method in this category is entropy query-by-bagging (EQB) [10]. MVSC, originating from the co-training technique, extends the SVSC AL into the MV scenario. Di and Crawford [37] first introduced this idea for hyperspectral image classification. Most MV AL approaches belong to this category, and SVM is usually adopted as the back-end classifier as it has demonstrated promising capabilities in the classification of remote sensing data [37]–[39]. Although these three AL frameworks have been demonstrated to be effective for hyperspectral image classification in the current literature, an MV ensemble classifier (MVEnC) AL framework, which may further improve the performance of MVSC, has not been explored in this field. Different with MVSC, which uses only one classifier for each

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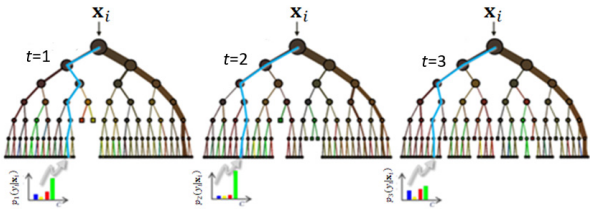


Fig. 2. Illustration of the forest class posterior probability [42].

view, MVEnC integrates a group of classifiers in each view to further improve the sufficiency of each view. This is reasonable because the ensemble classifier has already received significant attention due to its potential to significantly improve the classification accuracy of hyperspectral imagery.

Among the various ensemble techniques, RF is the most intuitive ensemble learning technique for the classification of high-dimensional data [41]. In RF, each tree is trained on a bootstrapped sample of the original training set, and the number of bootstrapped samples is the number of trees in the forest. The RF classifier produces a distribution that characterizes membership over the entire range of classes. In fact, each tree leaf stores the posterior probabilities for all the classes, which is the fraction of observations of a specific class in a tree leaf. During prediction, for any test sample $\mathbf{x}_i$, each tree leaf yields the posterior probability for class y_i, denoted as $p_t(y_i|\mathbf{x}_i)$. The final class distribution averages these probabilities over all the N trees in the ensemble and is defined as

$$p(y_i|\mathbf{x}_i) = \frac{1}{N}\sum_{t=1}^{N} p_t(y_i|\mathbf{x}_i), \quad y_i \in \{1, 2, \ldots, W\}. \tag{1}$$

This is also illustrated with a three-tree forest in Fig. 2 [42]. Fig. 2 also shows that a test sample $\mathbf{x}_i$ is sent through each tree to the appropriate child. The process is repeated until a leaf is reached, and the posterior probability $p_t(y_i|\mathbf{x}_i)$ from tree t is obtained. Averaging many tree posteriors can also help reduce the effect of possible noisy tree contributions.

C. Dynamic View Generation

To avoid all the views agreeing with each other as AL proceeds, we propose to construct a dynamic view at each AL iteration without changing the predefined fixed views. Unlike the more common MVSC-AL framework, RF is used as the back-end classifier in this MVEnC-AL approach. Besides its advantage of improving each view's performance, RF also has the ability to select features by analyzing the FI. For each tree in the forest, the associated out-of-bag (OOB) samples are the ones not included in the bootstrap samples used to construct this tree. Since the OOB samples are not used during the training stage, they can be used as the validation data to evaluate the trees' performances. Denote OOB_t as the out-of-bag samples for the tree t, and e_{OOB_t} as the classification error of the single tree t on the OOB_t data, and $\mathbf{x}^j$ as the vector of values associated with the jth feature and spanning all the samples in OOB_t. Then, randomly permute the values of $\mathbf{x}^j$ in OOB_t to obtain samples denoted as OOB'^j_t and compute the error of tree t on the permuted samples denoted as $\tilde{e}_{\text{OOB}'^j_t}$. The FI score associated with $\mathbf{x}^j$ is calculated as the mean error increase in the OOB samples, and it is equal to [43]

$$FI(\mathbf{x}^j) = \frac{1}{N}\sum_{t=1}^{N}(\tilde{e}_{\text{OOB}'^j_t} - e_{\text{OOB}_t}) \tag{2}$$

where N denotes the total number of trees in the RF. The larger the value of $FI(\mathbf{x}^j)$, the greater the importance of the feature $\mathbf{x}^j$.

Therefore, besides the fixed spectral–spatial views, a dynamic view is generated by stacking the most important features selected from the fixed views during the AL process. Since the FI changes as more training samples are added, the dynamic view is updated at each AL iteration and thus provides necessary disagreement information for the AL query.

D. AL Query and Decision Fusion

To enrich the training set during AL, we consider the maximum disagreement criterion, the capabilities of which have been demonstrated in identifying the most informative samples across MVs [37]. More formally, let $\hat{y}_i^{V_j}$ be the predicted label for pixel $\mathbf{x}_i$ provided by the back-end classifier from view V_j, and then, the uncertainty level of $\mathbf{x}_i$ can be expressed as

$$D_{unc}(\mathbf{x}_i) = |\hat{y}_i^{V_j}|_{j=1}^{J} \tag{3}$$

where J is the number of views and $|\cdot|$ counts the number of different estimates over J views.

After the learning is accomplished, a decision fusion strategy is applied to perform classification for each pixel. In this article, the individual posterior probability $p_j(y_i|\mathbf{x}_i)$ of each classifier is considered for making the final classification decision using the LOP method

$$P(y_i|x) = \sum_{j=1}^{J} a_j p_j(y_i|\mathbf{x}_i) \tag{4}$$

$$w = \arg\max_{y_i \in \{1,\ldots,W\}} P(y_i|x) \tag{5}$$

where W is the number of classes and $a_j (j = 1, 2, \ldots, J)$ is a classifier-dependent weight, which can either be assigned based on the "confidence score" of each classifier or can be uniformly distributed over all classifiers. In this article, we adopt the uniformly distributed weight for each classifier.

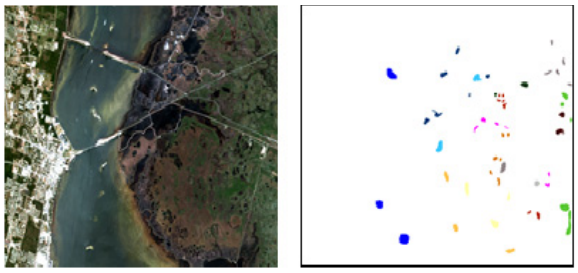
IV. Experiments

A. Data Set Description

The proposed MV AL framework was evaluated experimentally on three widely analyzed hyperspectral remote sensing data sets: Kennedy Space Center (KSC), Indian Pine, and University of Houston (UH).

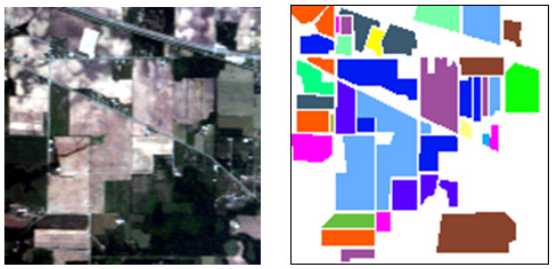
The KSC data set was acquired by the NASA Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the KSC in March 1996, with 18-m spatial resolution and 10-nm spectral resolution over the range of 400–2500 nm. The data cover a natural wetland/upland environment. Removal

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(a) (b)

Fig. 3. KSC data set. (a) True-color composite (bands R: 25, G: 13, and B: 7). (b) Ground reference map.



(a) (b)

Fig. 4. Indian Pine data set. (a) True-color composite (bands R: 26, G: 14, and B: 8). (b) Ground reference map.

TABLE I

NUMBER OF LABELED SAMPLES FOR THE KSC DATA SET

Class name	# labeled samples
C1: Scrub	761
C2: Willow swamp	243
C3: Cabbage palm hammock	256
C4: Cabbage palm/Oak	252
C5: Slash pine	161
C6: Oak/Broadleaf hammock	229
C7: Hardwood swamp	105
C8: Graminoid marsh	431
C9: Spartina marsh	520
C10: Cattail marsh	404
C11: Salt marsh	419
C12: Mud flats	503
C13: Water	927
Total	5211

TABLE II

NUMBER OF LABELED SAMPLES FOR THE INDIAN PINE DATA SET

Class name	# labeled samples
C1: Corn-no till	1434
C2: Corn-min till	834
C3: Corn	234
C4: Soybeans-no till	968
C5: Soybeans-min till	2468
C6: Soybeans-clean till	614
C7: Alfalfa	54
C8: Grass/Pasture	497
C9: Grass/Trees	747
C10: Grass/Pasture-mowed	26
C11: Hay-windrowed	489
C12: Oats	20
C13: Wheat	212
C14: Woods	1294
C15: Building-Grass-Trees-Drive	380
C16: Stone-Steel towers	95
Total	10366

TABLE III

NUMBER OF LABELED SAMPLES FOR THE UH DATA SET

Class name	# training samples	# testing samples
C1: Grass healthy	198	1053
C2: Grass stressed	190	1064
C3: Grass synthetic	192	505
C4: Tree	188	1056
C5: Soil	186	1056
C6: Water	182	143
C7: Residential	196	1072
C8: Commercial	191	1053
C9: Road	193	1059
C10: Highway	191	1036
C11: Railway	181	1054
C12: Parking lot 1	192	1041
C13: Parking lot 2	184	285
C14: Tennis court	181	247
C15: Running track	187	473
Total	2832	12197

of noisy and water absorption bands results in 176 bands. Ground reference information was composed of 5211 labeled samples associated with 13 classes: *Cabbage palm hammock*, *Cabbage palm/Oak*, *Cattail marsh*, *Graminoid marsh*, *Hardwood swamp*, *Mud flats*, *Oak/Broadleaf hammock*, *Salt marsh*, *Scrub*, *Slash pine*, *Spartina marsh*, *Water*, and *Willow swamp*. True-color composite and related ground reference maps are shown in Fig. 3, and the number of class-dependent labeled samples is shown in Table I.

The Indian Pine data set was acquired by the AVIRIS sensor over the Indian Pine agricultural site in Northwestern Indiana in June 1992, at 20-m spatial resolution and 10-nm spectral resolution over the range of 400–2500 nm. Most classes are crops that are located in fields with regular boundaries, resulting in a spatially structured data set. Twenty water absorption bands (104–108, 150–163, and 220) were removed, resulting in a 200-band image. Ground reference data were comprised of 10 366 labeled samples assigned to 16 classes: *Corn-no till*, *Corn-min till*, *Corn*, *Soybeans-no till*, *Soybeans-min till*, *Soybeans-clean till*, *Alfalfa*, *Grass/Pasture*, *Grass/Trees*, *Grass/Pasture-mowed*, *Hay-windrowed*, *Oats*, *Wheat*, *Woods*, *Building-Grass-Trees-Drives*, and *Stone-Steel towers*. True-color composite and related ground reference maps are shown in Fig. 4, while the details on the number of labeled samples are listed in Table II.

The UH data set [45] was acquired by the ITRES-CASI 1500 sensor over the UH campus and the neighboring urban area on June 23, 2012, between 17:37:10 and 17:39:50 UTC. The average height of the sensor was 5500 ft above ground, which resulted in 2.5-m spatial resolution data. The imagery consisted of 144 spectral bands ranging from 380 to 1050 nm and was processed (radiometric correction, attitude processing, GPS processing, geo-correction, and so on) to yield the final geocorrected image cube representing at-sensor spectral radiance, SRU $= \mu$W/cm^2 sr nm. The study area represented urban scenarios composed of 15 mutually exclusive classes. The classes were identified by photointerpreting a very-high-resolution optical imagery of the scene acquired during the same collection. Ground-truth reference information was subdivided into distinct and spatially disjoint subsets of training and testing data, as shown in Fig. 5. The details on the number of labeled samples are listed in Table III. We note that although the original data set contained also LiDAR data, only the hyperspectral imagery was considered in the context of this article.

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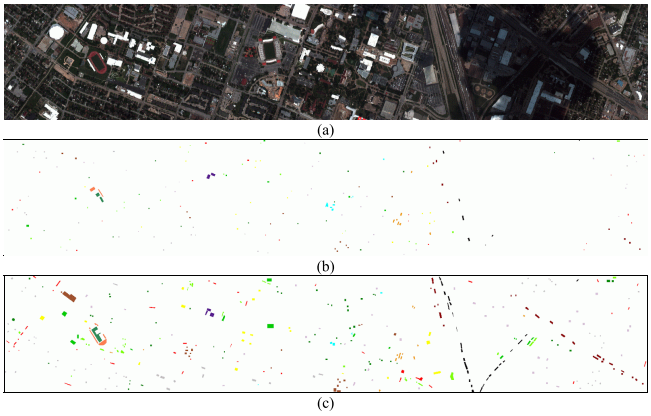


Fig. 5. UH data set. (a) True-color composite (wavelengths R: 650.2 nm, G: 531.1 nm, and B: 473.9 nm). Ground reference map for (b) training/learning set and (c) testing set.

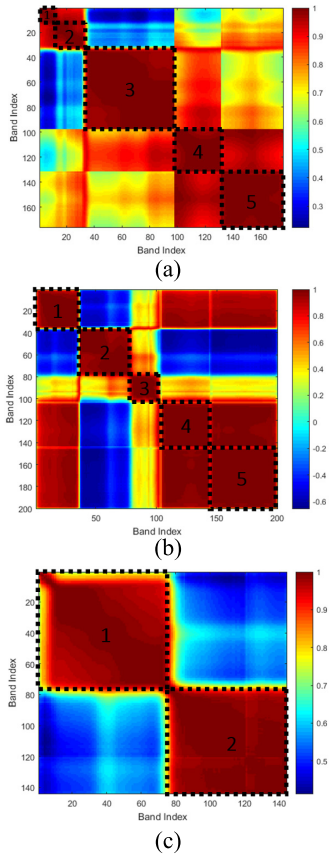


Fig. 6. Correlation matrix for (a) KSC, (b) Indian Pine, and (c) UH data sets.

TABLE IV
SPECTRAL CHARACTERISTICS OF EACH VIEW

KSC Dataset

View	Band Indices	Wavelength Range (nm)	Spectral Range
1	1 – 11	409.21 – 507.74	Blue
2	12 – 31	517.60 – 683.30	Green + Red
3	32 – 96	692.88 – 1284.20	Red + NIR
4	97 – 130	1294.15 – 1772.06	NIR + SWIR
5	131 – 176	1782.02 – 2437.40	SWIR

Indian Pine Dataset

View	Band Indices	Wavelength Range (nm)	Spectral Range
1	1 – 36	400.02 – 706.19	Blue + Green + Red
2	37 – 79	715.83 – 1119.66	Red + NIR
3	80 – 103	1129.25 – 1322.81	NIR
4	104 – 145	1382.54 – 1789.40	NIR + SWIR
5	146 – 200	1903.26 – 2498.96	SWIR

UH Dataset

View	Band Indices	Wavelength Range (nm)	Spectral Range
1	1 – 75	380 – 727	Blue + Green + Red
2	76 – 144	732 – 1050	Red + NIR

Blue: 435 – 500 nm; **Green**: 520 – 565 nm; **Red**: 625 – 750 nm
Near Infrared (NIR): 750 – 1500 nm
Short-wave IR (SWIR): 1500 – 3000 nm

B. Experimental Setup

Five spectral views were generated according to the correlation coefficient matrices for the KSC and the Indian Pine data sets, while only two views were considered for the UH data set (see Fig. 6). The band indices and spectral characteristics for all the views are summarized in Table IV.

The HSeg algorithm was applied to the full set of spectral bands as well as for the subsets of bands associated with the fixed views for each data set. Adopting the supervised pruning strategy proposed in [22], the best segmentation maps for the complete data sets and the subsets were selected from their corresponding HSegs with the initial training samples. The initial training set was chosen by selecting randomly five samples per class from the learning set, which resulted in 65, 80, and 75 samples for the KSC, the Indian Pine, and the UH data set, respectively. The AL algorithm was run until reaching 250 labeled samples, by adding ten labeled samples at each iteration. The entire procedure was repeated ten times, each time with a different randomly selected initial training set.

As noted previously, there are three components of the contribution of the proposed framework: 1) spectral–spatial view generation; 2) MV ensemble framework; and 3) view updating strategy. To analyze the contribution of each component of the proposed framework, the experimental analysis was conducted in three stages. Considering that the correlation-partition-based clustering method [37] which solely depends on the spectral information suffers from the insufficiency of the generated views, the spectral–spatial view generation strategy is first investigated. Then, to demonstrate the MVEnC framework as well as the view updating strategy, the proposed adaptive MVEnC AL approach (AdMVEnC_Spec+HSeg) was compared to: 1) fixed multiview with a single back-end classifier (MVSC_Spec+HSeg) and 2) fixed multiview with an ensemble back-end classifier (MVEnC_Spec+HSeg). Finally, to further demonstrate the proposed combined framework (AdMVEnC_Spec+HSeg), results were compared to those of five other baseline AL approaches, including: 1) SVSC based on pure spectral views (SVSC_Spec); 2) SVSC based on spectral–spatial views with spatial features extracted from HSeg (SVSC_Spec+HSeg); 3) fixed MVSC based on pure spectral views (MVSC_Spec); 4) fixed MVSC based on the spectral–spatial views with spatial features extracted from a fixed size window for each view (MVSC_Spec+Win); and 5) fixed MVSC based on spectral–spatial views with spatial features extracted from HSeg (MVSC_Spec+HSeg). A summary with the characteristics of the proposed and the comparison methods is given in Table V. In the comparison

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TABLE V
SUMMARY OF THE PROPOSED AND COMPARISON METHODS

Method	Added spatial features	Multi view	Classifier type
SVSV_Spec	No	No	SVM
SVSV_Spec+HSeg	Yes, from HSeg	No	SVM
MVSC_Spec	No	Yes, fixed	SVM
MVSC_Spec+Win	Yes, from windows	Yes, fixed	SVM
MVSC_Spec+HSeg	Yes, from HSeg	Yes, fixed	SVM
AdMVEnC_Spec+HSeg	Yes, from HSeg	Yes, fixed + adaptive	RF

approaches, SVM was adopted as the single back-end classifier. For the SVM, a Gaussian kernel was used, and the parameters C and γ for a Gaussian kernel were optimized in the range $[2^{-5}, 2^{15}]$ and $[2^{-15}, 2^{3}]$, respectively, every ten AL iterations of the process by a fivefold cross validation. In the proposed approaches, RF was adopted the ensemble back-end classifier, and the number of trees was set to 20. To generate the dynamic view, the 5, 20, and 10 most important features were selected from each view for the KSC, the Indian Pine, and the UH data set, respectively. These values were chosen automatically from the set [5, 10, 15, 20, 25, 30] using a cross-validation approach at the first AL iteration. In terms of the AL query, maximum disagreement was adopted for the MV-based approaches, and BT was adopted for the SV-based methods. Classification accuracies were evaluated in terms of multiple metrics: overall accuracy (OA), Kappa statistics, and class-specific accuracy. Results averaged on the ten independent runs were considered. Additionally, McNemar's test [44] was used to analyze the statistical significance of the results obtained by the different methods being compared. The test statistic is calculated as follows:

$$M = \frac{f_{12} - f_{21}}{\sqrt{f_{12} + f_{21}}} \tag{6}$$

where f_{12} is the number of samples correctly classified by the proposed method and misclassified by the method to which it is being compared, and f_{21} is defined analogously. The differences between the proposed method and others were assumed to be statistically significant at the 5% significance level if $|M| > 1.96$.

C. Experimental Results

1) KSC Data Set: For the first-stage evaluation, the classification accuracies of each view obtained by using the pure spectral features and the spectral–spatial features are shown in Fig. 7. To focus on the spectral–spatial view generation, only the fixed views were generated and the single SVM classifier was adopted as the back-end classifier in both cases. As expected, view 3 has the best performance over all views, because it lies in the red and NIR spectral range that is typically more useful to identify vegetation classes. Views 1 and 5 lie in the Blue and SWIR spectral range and have worse performance than other views. More importantly, the results show that adding spatial features into each view using the proposed method can significantly improve the classification accuracy of each view, which indicates the improvement of view sufficiency.

Fig. 8(a) and (b) shows the second-stage comparison results. For the ensemble versus single classifier comparison, it is

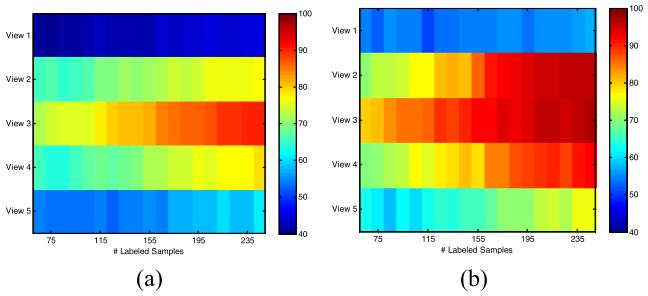


Fig. 7. Classification accuracies for the KSC data set of MVs based on (a) pure spectral views and (b) spectral–spatial views.

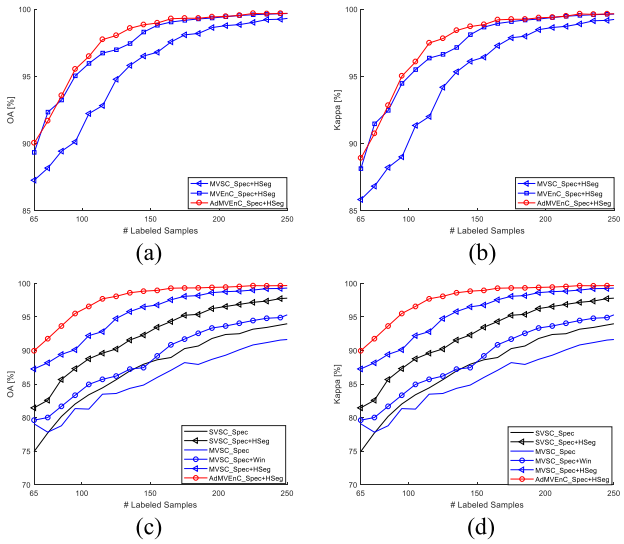


Fig. 8. OA and Kappa achieved on the KSC data set as a function of the number of labeled samples (a) and (b) proposed methods and (c) and (d) comparison methods.

evident that adopting the RF as the MV back-end classifier is much better than using the SVM, as all the ensemble classifier-based approaches have much higher accuracies than MVSC_Spec+HSeg through the entire AL process. In particular, using MVSC_Spec+HSeg in this MV setting, around 99% OA was reached at the very end. However, this accuracy was achieved by selecting fewer than 150 training samples with the proposed framework (AdMVEnC_Spec+HSeg). Furthermore, regarding the dynamic view component, we can see that adding a dynamic view achieved better performance than using the purely fixed views, and *AdMVEnC_Spec+HSeg* achieved higher accuracies than MVEnC_Spec+HSeg before ∼100% accuracy was obtained, which demonstrates the necessity of updating the views.

For the third-stage evaluation, the comparison of the proposed MV AL framework with the other baseline approaches is reported in Fig. 8(c) and (d). First, from the learning curves, it is obvious to see that for both SV and MV methods, incorporation of the spatial features is useful in the process of AL for this data set. Then, comparing MVSC_Spec+Win with MVSC _Spec+HSeg, the segmentation-based spatial feature extraction strategy is demonstrated to be more effective than the window-based strategy in the MV setting. This is because,

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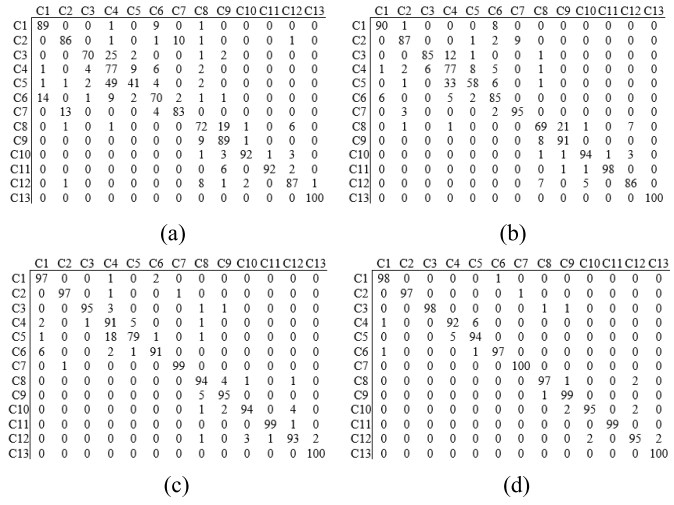


Fig. 9. Normalized confusion matrices (values in percentage) achieved on the KSC data set. (a) MVSC_Spec. (b) MVSC_Spec+Win. (c) MVSC_Spec+HSeg. (d) AdMVEnC_Spec+HSeg.

unlike the window-based strategy, a pixel has different neighborhoods in different views since HSeg was applied to each view independently, and therefore, the extracted spatial features can be more diverse. Finally, the results of SVSC_Spec and MVSC_Spec show that the MV is worse than the SV when using the pure spectral features, while the proposed method performed much better than all the SV-based approaches. Again, this confirms the effectiveness of adding the HSeg-based spatial features into each view, as it can provide the view-specific spatial information and therefore significantly improve the sufficiency of each view. Table VI shows the results from McNemar's test at the last AL iteration. The differences in classification accuracy between the proposed method and other baseline methods are statistically significant at the 5% level of significance. The class-specific accuracies averaged on the entire AL process are listed in Table VII for the comparison approaches. For some classes which are easy to discriminate, such as *scrub* (C1), *salt marsh* (C11), and *water* (C13), high accuracies were achieved for all the methods that include any form of spatial information. For the difficult classes, such as *slash pine* (C5) and *oak/broadleaf hammock* (C6), the proposed framework significantly outperforms other methods. Also, it is worth mentioning that AdMVEnC_Spec+HSeg achieved the highest accuracies in all the cases for this data set. To further examine the class-specific performance of the proposed method, we report the normalized confusion matrices (values in percentage) for the four MV-based approaches in Fig. 9. Again, the results clearly show that the hard classes are more likely to be confused with other classes. For example, *slash pine* (C5) is mainly confused with *cabbage palm/oak* (C4) in *MVSC_Spec* and MVSC_Spec+Win, and significant improvements were observed in the proposed method AdMVEnC_Spec+HSeg. Finally, we report the classification maps on the full scene in Fig. 10.

2) Indian Pine Data Set: The experimental results obtained on the Indian Pine data set confirmed the general trends

TABLE VI

RESULTS FROM MCNEMAR'S TEST FOR THE KSC DATA SET AT THE LAST AL ITERATION

Method	M
AdMVEnC Spec+HSeg vs. SVSV Spec	16.96
AdMVEnC Spec+HSeg vs. SVSV Spec+HSeg	9.07
AdMVEnC Spec+HSeg vs. MVSC Spec	20.22
AdMVEnC Spec+HSeg vs. MVSC Spec+Win	24.89
AdMVEnC Spec+HSeg vs. MVSC Spec+HSeg	3.60

TABLE VII

AAS ACHIEVED ON THE KSC DATA SET

	SVSC Spec	SVSC Spec+ HSeg	MVSC Spec	MVSC Spec+ Win	MVSC Spec+ HSeg	AdMVEnC Spec+ HSeg
OA	87.31	92.10	85.55	88.72	95.34	**97.63**
AA	81.99	89.54	80.46	85.93	94.01	**97.17**
Kappa	85.87	91.21	93.92	87.46	94.81	**97.36**
C1	90.75	94.25	88.55	90.48	96.80	**98.38**
C2	84.68	88.35	85.82	87.13	96.77	**97.49**
C3	80.96	90.50	69.81	85.31	94.80	**98.43**
C4	70.58	87.34	76.75	76.97	91.01	**92.30**
C5	51.40	72.27	40.95	58.49	79.00	**93.59**
C6	59.09	83.58	70.39	85.38	90.70	**97.48**
C7	80.94	89.66	82.87	94.71	98.71	**100**
C8	85.17	87.29	71.51	69.35	93.74	**96.86**
C9	90.18	91.57	89.33	91.11	94.90	**98.65**
C10	90.01	89.31	91.78	94.10	93.50	**95.35**
C11	91.44	97.39	91.58	97.90	98.93	**99.43**
C12	90.96	92.57	86.82	86.48	93.40	**95.25**
C13	99.77	99.98	99.82	99.62	99.89	**99.99**

obtained for the KSC data set. From Fig. 11, we note again that adding spatial features using the HSeg-based approach into each view increases the accuracy of each view greatly. It is related to the fact that the crops are actually quite small and the signatures are primarily soil and residue. Adding spatial information is quite useful in this case because it is the discriminating set of features.

Then, the results reported in Fig. 12(a) and (b) indicated clear advantages of using the ensemble back-end classifier. Focusing on the two ensemble classifier-based approaches, although the fixed view method MVEnC_Spec+HSeg is almost as good as the proposed method at the early stages, after several iterations, the fixed classifiers tend to agree with each other and no more information can be learned, and the accuracy tends to be stable without further improvement. In contrast, the dynamic views continue to learn as the AL proceeds.

In Fig. 12(c) and (d), the proposed framework was compared with the other baseline approaches. Also, in this case, the proposed framework (AdMVEnC_Spec+HSeg) yielded higher classification accuracies through the entire AL process. More specifically, without the spatial features, we can see that similar to the KSC data set, the MV method MVSC _Spec is worse than the SV method SVSC_Spec, and this is because using only the spectral features cannot guarantee the sufficiency of each view and combining the results from weak views could hamper the final performance. Again, for this data set, incorporating the spatial features is useful for both the SV and MV approaches. In the MV scenario, the HSeg-based spatial feature extraction strategy is more preferable than the window-based method, as the segmentation map extracted from HSeg is view-dependent, demonstrating the effectiveness of incorporating spatial features from diverse neighborhoods. The results from McNemar's test at the last

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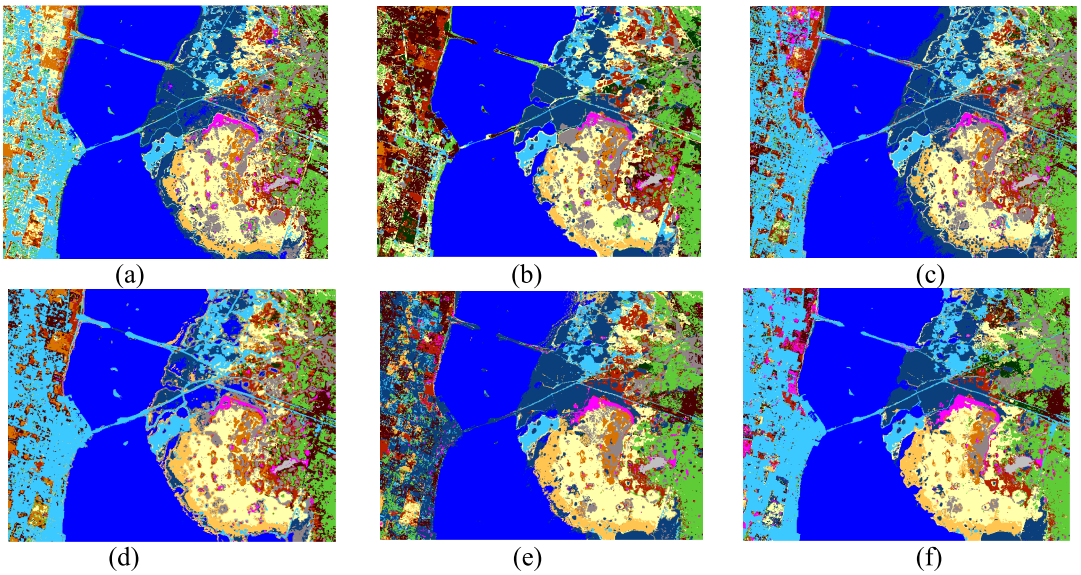


Fig. 10. Classification maps at the last iteration (averaged on the ten independent runs) for the KSC data set. (a) SVSC_Spec. (b) SVSC_Spec+HSeg. (c) MVSC_Spec. (d) MVSC_Spec+Win. (e) MVSC_Spec+HSeg. (f) AdMVEnC_Spec+HSeg.

TABLE VIII

RESULTS FROM THE MCNEMAR'S TEST FOR THE INDIAN PINE DATA SET AT THE LAST AL ITERATION

Method	M
AdMVEnC_Spec+HSeg vs. SVSC_Spec	40.07
AdMVEnC_Spec+HSeg vs. SVSC_Spec+HSeg	6.84
AdMVEnC_Spec+HSeg vs. MVSC_Spec	55.41
AdMVEnC_Spec+HSeg vs. MVSC_Spec+Win	48.54
AdMVEnC_Spec+HSeg vs. MVSC_Spec+HSeg	16.09

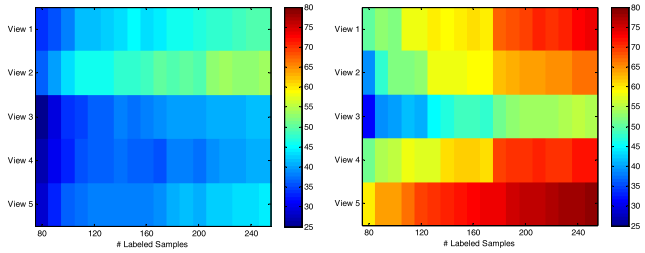


Fig. 11. Classification accuracies for the Indian Pine data set of MVs based on (a) pure spectral views and (b) spectral–spatial views.

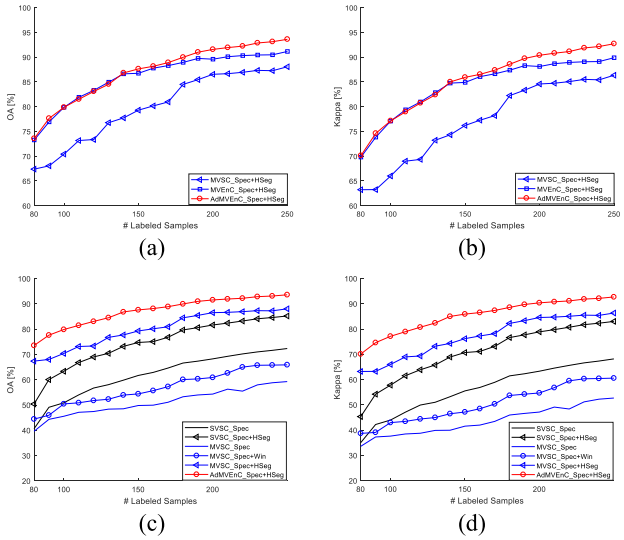


Fig. 12. OA and Kappa achieved on the Indian Pine data set as a function of the number of labeled samples (a) and (b) proposed methods and (c) and (d) comparison methods.

AL iteration are reported in Table VIII, and the differences between AdMVEnC_Spec+HSeg and other baseline methods are statistically significant. The overall and class-specific accuracies are summarized in Table IX, in which the proposed framework yielded the highest OA, average accuracy (AA), and most of the class-specific accuracies. We also report the normalized confusion matrices (values in percentage) for the four MV-based approaches in Fig. 13. For MVSC_Spec, C6: *soybeans-clean till* is most confused with C5: *soybeans-min till*, which differs only in the quantity of soybean residue remaining after tillage. Because tillage practices result in a range of values in the remaining residue, the class spectral values are not well clustered, and the accuracies based on class spectra are not surprising. C15: *building-grass-trees-drives* is a mixed class and is confused with classes containing its constituent components. Similar confusion matrices were obtained for the MVSC_Spec+Win and MVSC_Spec+HSeg methods [see Fig. 13(b) and (c)], while significant improvements were exhibited by the proposed framework [see Fig. 13(d)]. The classification maps on the whole scene are reported in Fig. 14.

3) UH Data Set: Finally, we validated the proposed methodology on the UH data set. Results in terms of OA, Kappa, and AA are reported in Fig. 15. The proposed method exhibited accuracies equal to 93.30%, 93.37%, and 92.82% in terms of OA, AA, and Kappa, respectively, at the last

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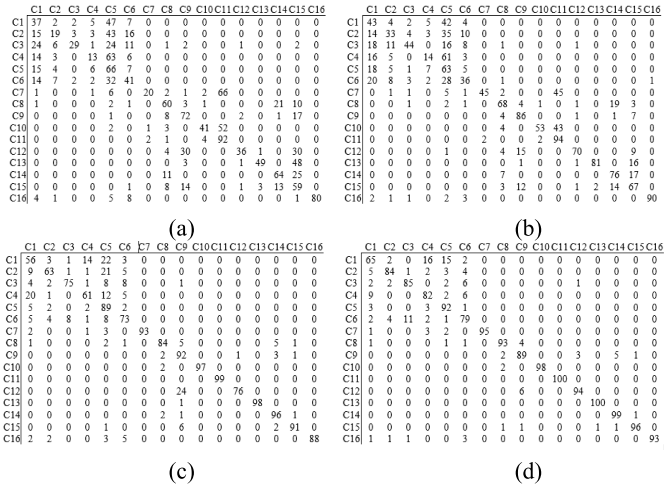


Fig. 13. Normalized confusion matrices (values in percentage) achieved on the Indian Pine data set. (a) MVSC_Spec. (b) MVSC_Spec+Win. (c) MVSC_Spec+HSeg. (d) AdMVEnC_Spec+HSeg.

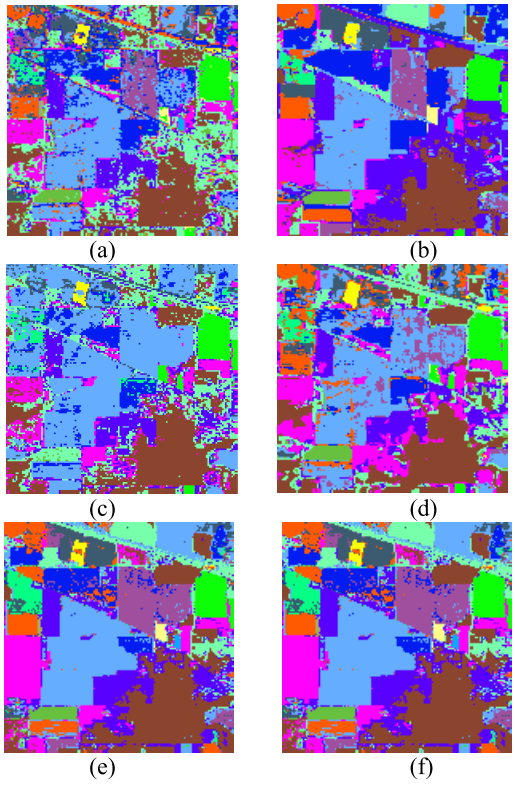


Fig. 14. Classification maps at the last iteration (averaged on the ten independent runs) for the Indian Pine data set. (a) SVSC_Spec. (b) SVSC_Spec+HSeg. (c) MVSC_Spec. (d) MVSC_Spec+Win. (e) MVSC_Spec+HSeg. (f) AdMVEnC_Spec+HSeg.

iteration when 245 samples were considered to train the classifier, which outperformed multiple state-of-the-art methods [46]–[49]. It is also important to note that the gain in terms of accuracies during the AL process was even greater than what was verified for the KSC and Indian Pine data sets when comparing the proposed methodology with respect to the baseline method based on purely spectral features (see Fig. 15). In more detail, for KSC, the difference in terms

TABLE IX

AAs ACHIEVED ON THE INDIAN PINE DATA SET

	SVSC Spec	SVSC Spec+ HSeg	MVSC Spec	MVSC Spec+ Win	MVSC Spec+ HSeg	AdMVEnC Spec+ HSeg
OA	61.96	81.64	51.19	56.87	80.00	**87.10**
AA	62.16	85.52	48.64	60.19	83.14	**90.13**
Kappa	56.30	79.11	43.67	50.36	77.07	**85.34**
C1	51.22	76.77	36.71	42.76	56.18	**64.59**
C2	33.82	72.94	19.12	32.68	63.01	**84.09**
C3	43.21	66.82	28.62	43.83	74.83	**84.95**
C4	49.94	59.24	13.08	14.10	60.83	**82.20**
C5	71.93	82.27	66.11	63.09	88.67	**92.08**
C6	31.99	75.26	41.49	36.37	72.81	**79.00**
C7	41.26	93.30	20.38	44.52	92.94	**94.58**
C8	59.80	86.59	59.88	67.93	83.62	**92.98**
C9	79.38	**94.45**	71.55	86.46	92.36	89.06
C10	86.00	96.94	41.37	52.80	97.27	**97.78**
C11	87.21	99.27	92.43	94.33	99.18	**99.71**
C12	77.25	**95.56**	35.58	69.89	75.94	94.36
C13	82.24	98.55	48.71	80.95	98.14	**99.56**
C14	85.49	91.13	63.54	75.80	96.21	**98.63**
C15	29.75	88.99	59.41	66.99	90.63	**95.65**
C16	84.05	89.74	80.22	90.48	87.56	**92.87**

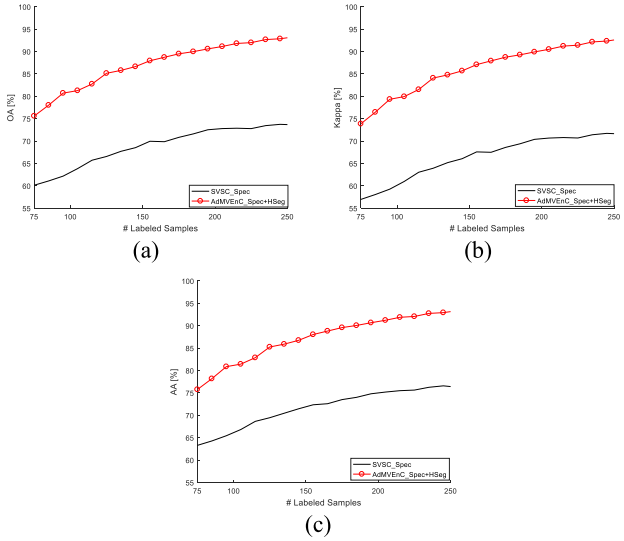


Fig. 15. (a) OA, (b) Kappa, and (c) AA achieved on the UH data set as a function of the number of labeled samples (average values from ten repeated runs).

of OA at the starting point between the proposed solution and the baseline method was around 15%, and remained roughly constant in the first several iterations and was then close to 5% at the last iteration. A similar pattern was observed for the Indian Pine data set, in which the gain in terms of OA was close to 30% in the first several iterations and close to 20% at the last one. For the UH data set, the OA difference at the starting point was close to 15% and increased to 21% at the last iteration.

4) Free Parameters Analysis: The proposed methodology has two free parameters, for which we conducted an extensive sensitivity analysis. The first one is the number of trees in the RF classifier. We varied this parameter in the set [10, 20, 50, 100] and evaluated the differences in accuracies for the KSC and the Indian Pine data sets, as reported in Fig.16 (a) and (b). As expected, an increase of this parameter brought an increase in the accuracy. However, such differences were quite limited (especially when a number of trees $\geq$ 20) and no obvious differences were verified at the last iteration of the AL process among the different tested parameter values. Considering that a larger number of trees are associated with a higher

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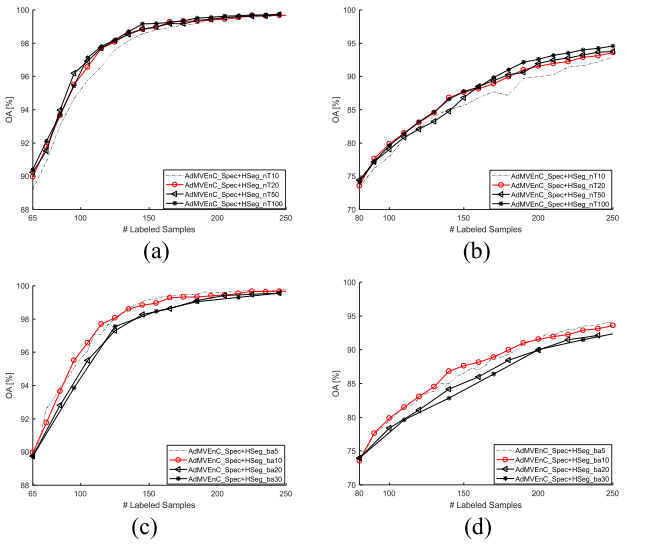


Fig. 16. Sensitivity to the free parameters of the proposed methodology. OA as a function of the number of labeled samples and at varying the number of trees (nT) in the RF classifier for (a) KSC and (b) Indian Pine and at varying the batch size (ba) in the AL selection process for (c) KSC and (d) Indian Pine.

computational load, we fixed the tree number to 20 for all the experiments.

The second free parameter is the batch size of the AL process, which we empirically set to 10 in all the experiments. This is a common parameter for any AL methods and for which no strategies exist to estimate it in an automatic way. In general, decreasing the batch size is associated with the accuracy improvement while increasing the total number of AL iterations. We varied this parameter in the set [5, 10, 20, 30] and evaluated the differences in accuracies for the KSC and Indian Pine data sets, as reported in Fig. 16(c) and (d). A decrease of the parameter brought an expected increase in the accuracy. However, also in this case, such differences were quite limited (especially when a batch size ≤ 10) and no obvious differences were found at the last iteration among the different compared values.

5) Computational Complexity Analysis: The effectiveness of the proposed framework was also analyzed in terms of computational complexity in comparison with the other AL approaches. The offline part consists of generating the predefined spectral–spatial views. Considering the independence of the fixed views, computational improvement can be achieved by calculating them simultaneously in parallel on different processors or clusters. For the online AL part, the computational time for the KSC data set is reported for the approaches with spatial features derived from HSeg in Fig. 17, including SVSC_Spec+HSeg, MVSC_Spec+HSeg, and AdMVEnC_Spec+HSeg, as they achieved higher classification accuracies than others based on either purely spectral features (SVSC_Spec and MVSC_Spec) or window-based spatial features. The results show that both SVSC_Spec+HSeg and MVSC_Spec+HSeg have discontinuities, since these two approaches applied SVM as the back-end classifier, and these "jumps" occur when the cross validation was applied. The proposed solution

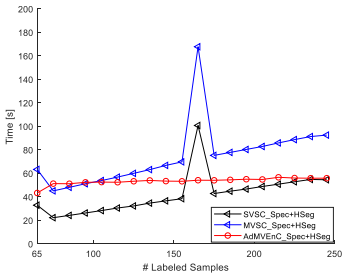


Fig. 17. Computational time as a function of the number of labeled samples for KSC data set (average values on ten independent runs).

that used RF as the back-end classifier consistently outperformed MVSC_Spec+HSeg. This is because SVM is time-consuming at the prediction phase, due to the computation of the kernel distances between each unlabeled sample and each support vector. This also indicates that traditional SVM-based AL approaches are problematic for large-scale data sets when the predictions for a large quantity of unlabeled samples are required. Additionally, although the two MV-based approaches, MVSC_Spec+HSeg and AdMVEnC_Spec+HSeg, need longer computational time compared to the SV approach SVSC_Spec+HSeg in the current sequential implementation, considering the independence of the predefined spectral–spatial views, significant computational improvement can be achieved by adopting the parallel computing strategy. Finally, considering the proposed framework can provide similar classification accuracies to other methods using much fewer labeled samples, the time spent on the human labeling can also be greatly reduced.

V. Conclusion

In this article, we have proposed a new MV AL framework for the classification of hyperspectral data with both spectral and spatial features. While traditional MV AL approaches for remote sensing data classification consider a single type of feature for view generation, and the views are mostly fixed, the proposed method combines spectral and spatial features in each view and is adaptive. The spatial features are extracted from HSegs, and an adaptive view is generated during the AL process by combining the important features from other fixed views. In addition, this is the first known introduction of an MVEnC in an AL framework to the remote sensing field. The ensemble back-end classifier is demonstrated to be more effective than the traditional single classifier.

The proposed framework was validated experimentally on three well-known hyperspectral data sets. The results show that incorporating spatial features during view generation can greatly enhance the view sufficiency, and the dynamic view helps promote the disagreement level as AL proceeds. Moreover, the proposed framework achieved significantly higher OAs and better performance for individual classes, relative to other baseline AL approaches in terms of the overall performance, demonstrating its value for selecting the best training data from existing labeled data sets and potentially reducing the manual labeling effort. Considering the independence of the fixed views, the proposed framework also has

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the potential to achieve lower computational complexity via parallel computing compared to SV-based methods. Further studies will be devoted in this direction.

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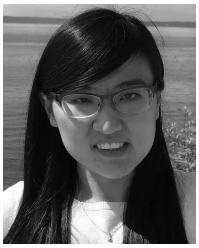
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Zhou Zhang (S'15–M'19) received the B.S. degree in astronautics engineering and the M.S. degree in instrumentation science and optoelectronics engineering from Beihang University, Beijing, China, in 2010 and 2013, respectively, and the Ph.D. degree from geomatics and civil engineering from Purdue University, West Lafayette, IN, USA, in 2017.

She is currently an Assistant Professor of biological systems engineering with the University of Wisconsin–Madison, Madison, WI, USA. Her research interests include multisource data fusion, UAV-based imaging platforms developments, hyperspectral image analysis, machine learning, and its related applications to remote sensing data.



Edoardo Pasolli (S'08–M'13–SM'17) received the M.Sc. degree in telecommunications engineering and the Ph.D. degree in ICT from the University of Trento, Trento, Italy, in 2008 and 2011, respectively.

He was a Post-Doctoral Fellow with the University of Trento from 2011 to 2012 and from 2014 to 2016, the NASA Goddard Space Flight Center, Greenbelt, MD, USA, from 2012 to 2013, and Purdue University, West Lafayette, IN, USA, from 2013 to 2014. He was a Marie Sklodowska-Curie Individual Fellow with the University of Trento from 2016 to 2018. Since December 2018, he has been an Assistant Professor with the Department of Agricultural Sciences, University of Naples Federico II, Naples, Italy. His research interests aim at developing machine learning and data science methodologies for complex ecosystems, including the processing of remote sensing data.



Melba M. Crawford (M'90–SM'05–F'08) is currently the Nancy Uridil and Francis Bossu Professor of civil engineering with Purdue University, West Lafayette, IN, USA, where she is also a Professor with the Schools of Civil Engineering and Electrical and Computer Engineering and the Department of Agronomy. She is co-leading a joint initiative between the Purdue Colleges of Agriculture and Engineering in the development of advanced sensing technologies and analysis methodology for wheeled and UAV platforms, focused on high-throughput phenotyping. Her research interests focus on advanced methods for image analysis, including dimensionality reduction, active learning, and deep learning for classification and prediction, and applications of these methods to hyperspectral and LIDAR data for agriculture and natural resource mapping and monitoring.

Dr. Crawford is the Past President of the IEEE Geoscience and Remote Sensing Society, an IEEE GRSS Distinguished Lecturer, and the current Treasurer of the IEEE Technical Activities Board. She was a member of the NASA EO-1 Science Validation Team and served on the NASA Earth System Science and Applications Advisory Committee and the Advisory Committee to the NASA Socioeconomic Applications and Data Center (SEDAC).

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